Enhanced Road Crack Detection

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I. Motivation

- ❑ Road damage impacts safety and infrastructure maintenance
- ❑ Manual inspection is costly and not scalable
- ❑ Dashcam data enables large-scale, real-world road monitoring

Goal: Enhance automated road crack detection using dashcam data



II. Related Work

Existing Approaches:
- ❑ **DeepCrack**: captures fine crack patterns using multi-scale features.[1]
- ❑ **YOLO-based models**: achieve high accuracy on curated datasets. [2]
- ❑ **CNN classifiers**: effective for labeled pavement dataset.[3]

Key Limitations:
- ❑ Struggle in **real-world conditions** (noise, lighting, occlusion)
- ❑ Depend on **pre-cropped or clean road images**
- ❑ Limited ability to handle **complex backgrounds and perspectives**



III. Proposed Method



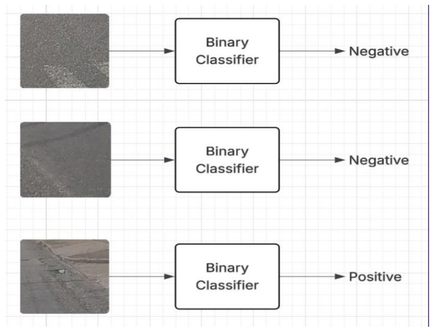
Dashcam frame extraction: Captures real-world road conditions for scalable monitoring



Road Segmentation, High Frequency Component: The method first focuses only on the road by removing surrounding background from the dashcam image. It then highlights important details on the road surface



Grid Based Patch Extraction: The road mask and high-frequency (HFC) image are combined to retain only crack-relevant edges while reducing background noise. The filtered image is then divided into smaller grid patches to enable localized analysis. Patches with strong edge patterns are selected as candidate regions for crack detection.[4]



Binary Classifier: Filtered regions may still contain edges or marks that look like cracks. A binary classifier is then used to determine whether each patch contains a crack or not.

IV. Results

Data Description

Public1: A balanced dataset with labeled crack and non-crack images used for initial model training and evaluation[4]
Public2: A larger dataset used to further improve the model by providing more diverse crack examples
UW Dataset: A real-world dashcam dataset collected locally to test model performance in practical road conditions

Experiments

Model 1: Base Training: The model is trained on Public1 and achieves strong performance on similar (in-domain) data.[4]

Model 2: Fine-Tuned ResNet: Fine-tuning on Public2 improves performance on UW data, achieving **~74% accuracy** with more balanced predictions.

Baseline Comparison (ResNet vs Qwen): Fine-tuned ResNet outperforms Qwen (74% vs 71%) with more balanced precision and recall, while Qwen shows higher recall but more false positives

Results ***FT= Fine Tuning***

Model	Accuracy	Precision	Recall	F1 Score
ResNet Base Training	0.56	0.61	0.56	0.51
Qwen Baseline	0.71	0.64	0.96	0.77
ResNet Head FT	0.74	0.74	0.74	0.74
ResNet Full FT	0.78	0.77	0.80	0.70

V. Conclusions

- ❑ While the model achieves high accuracy on Public1, its performance drops on real-world UW data, highlighting the impact of domain differences
- ❑ Fine-tuning improves results, but there is still a gap compared to in-domain performance
- ❑ In the future, using more real-world data and clustering similar crack patterns could make the model more reliable without heavy annotation

References

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